

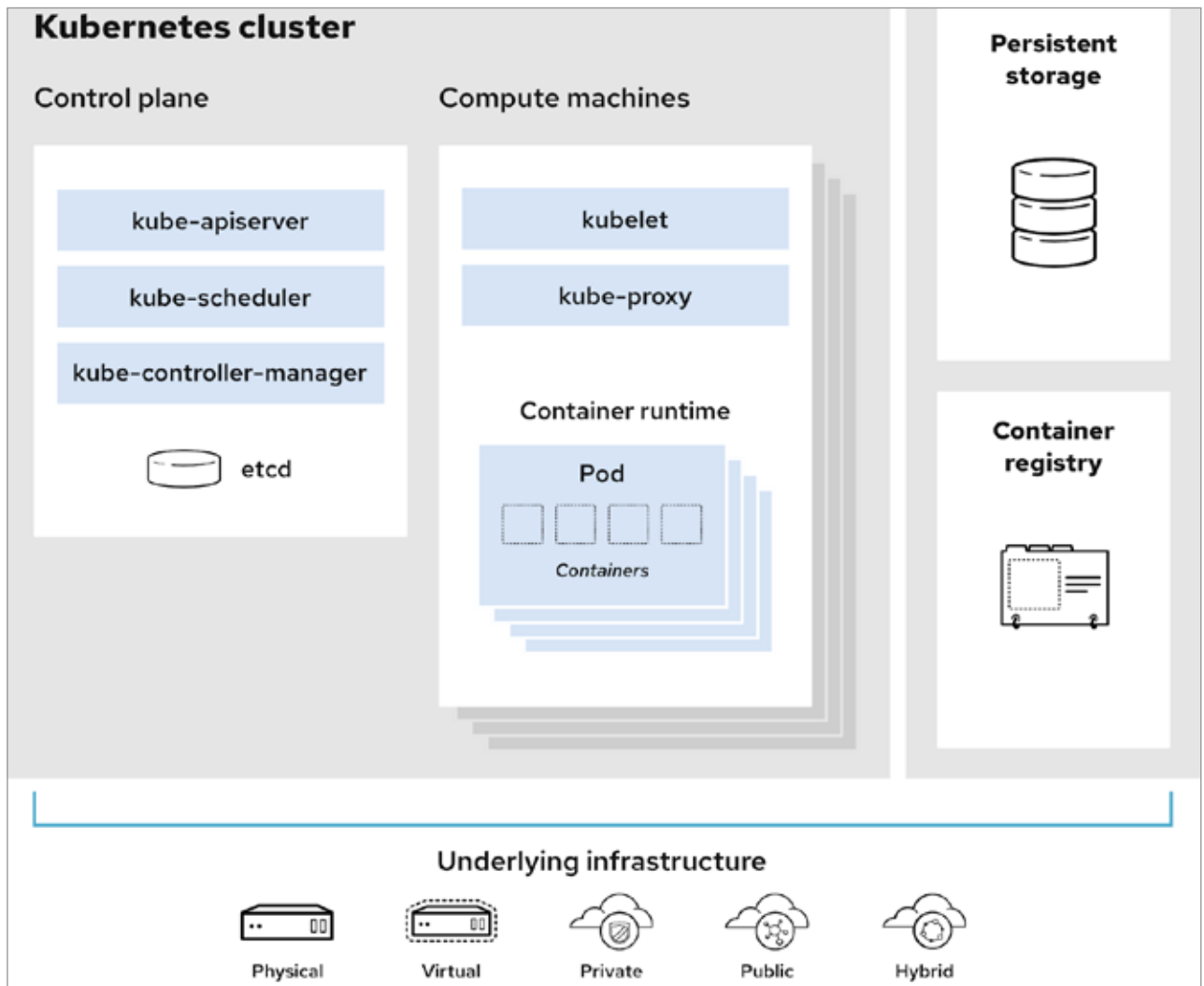


Figure 1: Kubernetes architecture (Credit: <https://www.redhat.com/en/topics/containers/what-is-kubernetes>)

- **Kubelet:** As an agent, Kubelet ensures the necessary containers are actively running within a Kubernetes pod.
 - **Kube-proxy:** This network proxy, operating on each node in the cluster, upholds network rules and facilitates communication.
 - **Container runtime:** This is the application that manages the operation of containers. Any runtime that complies with the Kubernetes CRI (container runtime interface) is supported by Kubernetes.
- Kubernetes is used to create applications that are easy to manage and deploy anywhere. When available as a managed service, Kubernetes offers a range of solutions to meet your needs. Here are some major use cases of Kubernetes.
- **Increasing development velocity:** Kubernetes helps to build cloud-native microservices-based apps. It also supports containerising existing apps, becoming the foundation of application modernisation and letting you develop apps faster.
 - **Deploying applications anywhere:** Kubernetes is built to be used anywhere, allowing you to run your applications across on-site public clouds and hybrid deployments. So you can run your applications where you need them.
 - **Running efficient services:** Kubernetes can automatically adjust the cluster size required to run a service. It enables you to automatically scale your applications up and down, based on the demand, and run them efficiently.

Why Kubernetes?

In a production scenario, continuous availability and efficient resource utilisation are critical. Kubernetes addresses these needs by automating the deployment, scaling, and management of containerised applications.